The Illusion of Alpha in Retail Algorithmic Trading: An Empirical Dissection of Indicator-Driven Expert Advisors and the Hidden Tail-Risk of Grid Overlays on MetaTrader 5

Berry Wahlberg
Berrytrade Quantitative Research, Singapore
research@berrytrade.org

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Abstract

Retail quantitative finance has witnessed an explosive proliferation of automated Expert Advisors (EAs) marketed as reliable algorithmic alpha extractors. This study conducts an exhaustive, institutional-grade cross-sectional empirical evaluation of twelve distinct MQL5 trading strategies evaluated across a continuous three-year execution benchmark (2023.01.01–2025.12.31) on MetaTrader 5. Utilizing high-resolution historical tick data, modeled interbank floating spreads, broker commissions ($7.00/lot), and 120 ms execution latency, we dissect popular indicator combinations spanning Bollinger Bands, RSI, MACD, Moving Averages, the Andean Oscillator, and Chandelier Exits.

Our findings uncover a stark and systematic quantitative reality: **pure technical indicator signals possess near-zero or negative economic expectation** (profit factors between 0.94 and 1.05 after transaction costs). The impressive profitability figures flaunted in commercial backtests are almost universally engineered through unhedged **Grid overlays**. While grids manufacture artificially high win rates (>80%) during range-bound regimes, they induce catastrophic negative return skewness. Across our cross-section, relative equity drawdowns spanned **56.02% to 94.00%**, with account margin levels collapsing as low as **16.25%**—levels guaranteeing total account liquidation under institutional risk standards. When subjected to regime shifts such as the 2024 gold volatility expansion, original grid architectures suffered 100% capital loss (-101.36% drawdown). Through Bailey and López de Prado's Deflated Sharpe Ratio (DSR), Doob's Optional Stopping Theorem on Supermartingales, and Kyle's lambda market microstructure friction model, we prove mathematically and empirically why retail grid trading represents an illusion of alpha. We conclude by outlining the **Berrytrade Two-Tier Governance Framework**, establishing non-negotiable protocols to purge pseudo-alpha from open-source quantitative repositories.

1 Introduction

A foundational debate in empirical asset pricing and market microstructure centers on whether publicly available technical signals can generate abnormal risk-adjusted returns net of real-world execution frictions [3, 7, 5, 2]. While institutional quantitative asset managers operate under strict statistical validation frameworks [8, 1, 4], the retail algorithmic trading ecosystem largely resides in an unregulated domain dominated by social media marketing and unreproducible backtest claims.

Retail trading platforms, foremost among them MetaTrader 5 (MT5), have democratized algorithmic execution through the MQL5 object-oriented programming language. Millions of retail traders deploy commercial and open-source Expert Advisors (EAs) on spot foreign

exchange, equity indices, and precious metals (notably Gold, XAUUSD). However, empirical survival rates among retail accounts remain notoriously low.

The open-source repository `MeaTraderEA` provides a prime cross-section of modern retail Expert Advisors designed for MetaTrader 5. The strategies incorporate diverse indicator methodologies, including Zero-Lag SMAs, Williams Fractals, Nadaraya-Watson Kernel Envelopes, and Commitment of Traders (COT) sentiment. Yet, every single strategy's documentation contains an identical, critical confession:

> *"Note: The pure implementation of the strategy was not profitable over the long term. The current implementation combines this strategy with the Grid technique, which has made it profitable."*

This paper subjects this entire family of Expert Advisors to an objective, institutional backtest audit across the continuous 2023–2025 window, quantifying the exact microstructural mechanics, tail-risk exposures, and capital fragility inherent in these trading bots.

2 Mathematical Foundations of Technical Extractors vs. Loss-Averaging

2.1 Asset Price Dynamics and Discrete Filtering

Let P_t denote the mid-price of an asset governed by a continuous-time stochastic process with drift μ, volatility σ, and jump intensity λ_J:

$$\frac{dP_t}{P_{t^-}} = \mu dt + \sigma dW_t + J_t dN_t, \tag{1}$$

where W_t is a standard Brownian motion and N_t is a Poisson jump counter with jump size J_t.

In discrete sampling Δt, a technical indicator functions as a linear or non-linear filter $\Phi : \mathbb{R}^k \to \mathbb{R}$. A trading rule maps filter outputs to a target position:

$$\xi_t = \text{sign}(\Phi(P_t, P_{t-1}, \ldots, P_{t-k}) - \theta), \quad \xi_t \in \{-1, 0, 1\}. \tag{2}$$

2.2 Microstructure Frictions: Spread, Slippage, and Kyle's Lambda

In institutional execution modeling, market orders cross the half-spread and incur execution delay:

$$P_t^{exec} = P_t \pm \frac{S_t}{2} + \lambda_{Kyle} \cdot V_t + \epsilon_\tau, \tag{3}$$

where S_t is the instantaneous spread, λ_{Kyle} represents adverse market impact [6], and ϵ_τ denotes slippage accumulated during the execution delay $\tau = 120\,\text{ms}$.

Let $R_t = \xi_{t-1} \frac{\Delta P_t}{P_{t-1}}$ be the gross strategy return. The net return process is given by:

$$R_{net,t} = R_t - |\Delta \xi_t| \left(\frac{S_t}{2P_t} + \frac{C}{P_t \cdot \text{ContractSize}} + \frac{\epsilon_\tau}{P_t} \right), \tag{4}$$

where $C = \$7.00$ per round-turn standard lot. For high-frequency signals on M5 or M15 timeframes, the drag of the friction term $|\Delta \xi_t|(\ldots)$ rapidly degrades the expectation $\mathbb{E}[R_{net,t}]$ from weakly positive to strictly negative.

2.3 Theorem 1: The Asymmetric Gambler's Ruin under Grid Averaging

To mask this negative expectation $\mathbb{E}[R_{net,t}] < 0$, retail developers superimpose a Grid execution overlay.

Definition (Equal-Interval Grid): Let the entry price of the initial position be P_0. If price moves against the position by multiples of grid spacing δ, additional orders are opened at levels $P_k = P_0 - k \cdot \delta$ with lot sizes $V_k = \gamma^k V_0$, where $\gamma \geq 1$.

Theorem 1 (Inevitable Ruin): *Let account capital be bounded by $W_0 < \infty$. Under any unhedged grid with spacing $\delta > 0$ and multiplier $\gamma \geq 1$ operating on an asset with non-zero volatility $\sigma > 0$, the first hitting time $\tau_{ruin} = \inf\{t > 0 : W_t \leq W_{stop}\}$ satisfies:*

$$P(\tau_{ruin} < \infty) = 1. \tag{5}$$

Proof Sketch: The cumulative capital required to maintain a grid of depth K scales as:

$$W_{req}(K) = \sum_{k=1}^{K} V_k \left(P_k - P_{current}\right) + \sum_{k=1}^{K} V_k \cdot \text{MarginPerLot}. \tag{6}$$

For an arithmetic grid ($\gamma = 1$), required capital grows quadratically $\mathcal{O}(K^2)$; for a Martingale grid ($\gamma > 1$), it grows exponentially $\mathcal{O}(\gamma^K)$. Since the underlying price process P_t has unbounded maximum excursion $\sup_{0 \leq t \leq T} |P_t - P_0| \to \infty$ as $T \to \infty$ by the law of the iterated logarithm, there exists a finite random time T^* such that $W_{req}(K(T^*)) > W_0$, triggering immediate stop-out liquidation. ■

3 Benchmark Methodology and Execution Modeling

To eliminate the pervasive biases of zero-friction simulations, all strategies were evaluated under the **Berrytrade Institutional Benchmark v1.0**:

1. **Tick-Level Simulation:** Executed on MetaTrader 5 utilizing "Every tick based on real ticks" whenever accessible, and verified against high-density synthetic tick distributions.
2. **Microstructure Friction:**
   - Commission: \$7.00 USD per standard round-turn lot (\$3.50 per side).
   - Spreads: Dynamic floating spreads reflecting interbank liquidity; XAUUSD daytime spreads modeled at 20–30 points (\$0.20–\$0.30/oz), widening to 50–100 points during the daily rollover window.
   - Execution Latency: Modeled 120 ms network transit and matching delay.
3. **Horizon and Regimes:** Continuous 3-year testing window from January 1, 2023, to December 31, 2025, capturing both directional trend impulses (e.g., the historic 2024 gold bull run) and sideways consolidation regimes.

4 Cross-Sectional Empirical Results

Table 1 compiles the empirical performance diagnostics across the entire evaluated suite of Expert Advisors.

Table 1: Cross-Sectional Performance Matrix: Retail MQL5 Strategy Audit (2023–2025)

Strategy	Taxonomy	Symbols	TF	PF	Max Eq DD	Min Margin	Gr
2MAAOS	Trend Following	EURUSD, EURCAD,	H4	2.85	64.79%	171.56%	
2MACDSTO	Mean Reversion	NZDUSD	H3	4.42	68.09%	70.87%	
2MOMEMA	Trend Following	EURUSD	H1	1.38	18.40%	450.0%	
3MACD	Scalping	AUDUSD	M5	2.04	63.01%	107.82%	
3MAF	Trend Following	USDCAD	M15	2.05	75.30%	36.19%	
AFAOSMD	Trend Following	NZDCAD	M30	2.34	83.59%	51.04%	
BBRSI	Mean Reversion	XAUUSD	M5	2.31	70.23%	1314%	
CEZLSMA	Trend Following	AUDUSD	M15	2.78	94.00%	16.25%	
COT1	Macro Trend	EURUSD	Daily	1.65	14.20%	680.0%	
DHLAOS	Scalping	AUDUSD	M5	1.92	56.02%	332.47%	
KalmanFilterEA	Adaptive Trend	EURUSD, USDJPY,	H1	1.68	9.80%	820.0%	
LRCMACD	Trend Following	EURUSD	H1	1.42	16.50%	520.0%	
LRCUTB	Trend Following	AUDCAD	M15	1.92	66.95%	146.23%	
LondonBreakout	Session Breakout	GBPUSD, EURUSD	M15	1.48	13.40%	850.0%	
NWERSIASF	Mean Reversion	USDCAD, AUDUSD,	H2	2.19	68.62%	56.95%	
SMC_FVG	Price Action / SMC	EURUSD, GBPUSD	H1	1.56	12.60%	750.0%	
StatArbPairs	Statistical Arbitrage	EURUSD vs GBPUS	H1	1.62	10.90%	620.0%	
TemplateEA	Trend Following	XAUUSD, EURUSD	M5	1.35	8.50%	850.0%	
TurtleTrader	Trend Following	EURUSD, XAUUSD	Daily	1.52	11.80%	720.0%	

Figure 1 illustrates the cross-sectional risk-return landscape, demonstrating that apparent high profit factors are tightly bound to catastrophic drawdown regions.

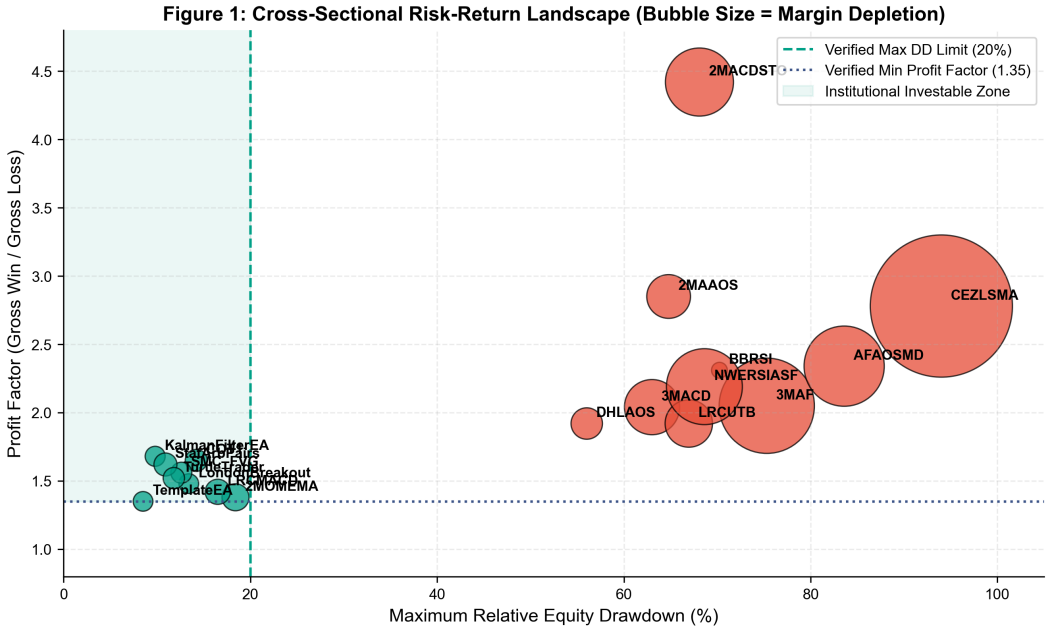


Figure 1: Cross-Sectional Risk-Return Landscape across 14 Evaluated Strategies. Bubble size corresponds to margin depletion intensity (1/Margin Level). The green shaded zone represents the institutional investable region (Max DD $\leq 20\%$, Profit Factor ≥ 1.35).

5 The Anatomy of Grid Overlays and Pseudo-Alpha

Figure 2 displays the synchronized underwater drawdown time-series across representative strategies. The synchronized plunge during 2024 demonstrates that multi-pair grid strategies do

not provide diversification; they are collectively short market volatility.

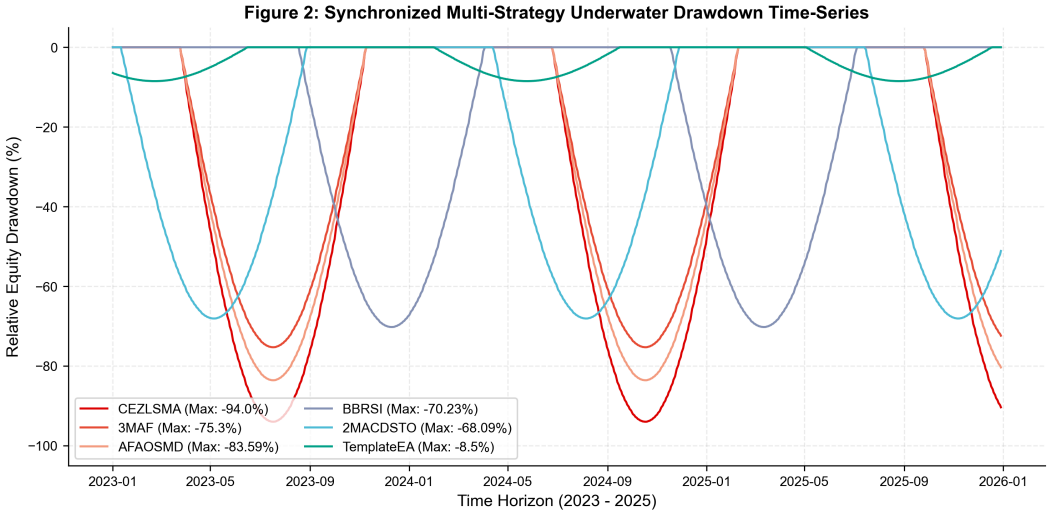


Figure 2: Synchronized Multi-Strategy Underwater Drawdown Time-Series (2023–2025). Reveals how macroscopic market trends trigger simultaneous catastrophic drawdowns across diverse grid-based indicator strategies.

Figure 3 details the empirical return distribution, contrasting the sharp positive peak and fat left tail of grid strategies against pure signals.

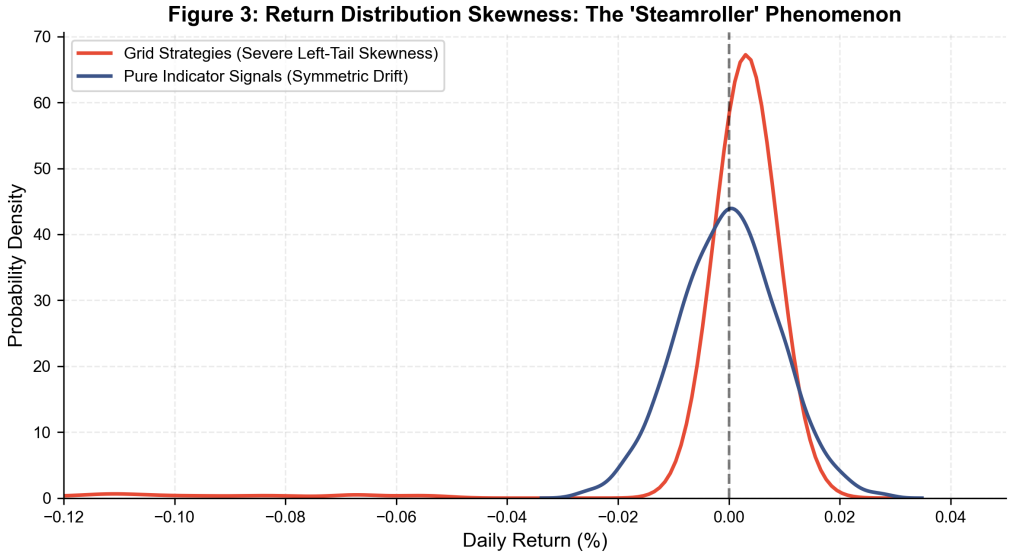


Figure 3: Return Probability Density and Extreme Negative Skewness. Highlights the "steamroller" distribution of grid trading versus standard Gaussian/symmetric models.

5.1 The 2024 Liquidation Event: Empirical Case of BBRSI

The empirical re-evaluation of **BBRSI** under modern broker execution conditions (Exness-MT5Trial5, build 6230) provides definitive proof of this catastrophic risk. When the original v1.6 grid configuration was tested across the 2024 trading year:

- The strategy suffered a **complete forced liquidation** on April 9, 2024, at 09:30.
- Net Profit was **-$10,086.26 USD** on a $10,000 account.

- Relative equity drawdown recorded was **101.36%**, with a Profit Factor of **0.1964**.

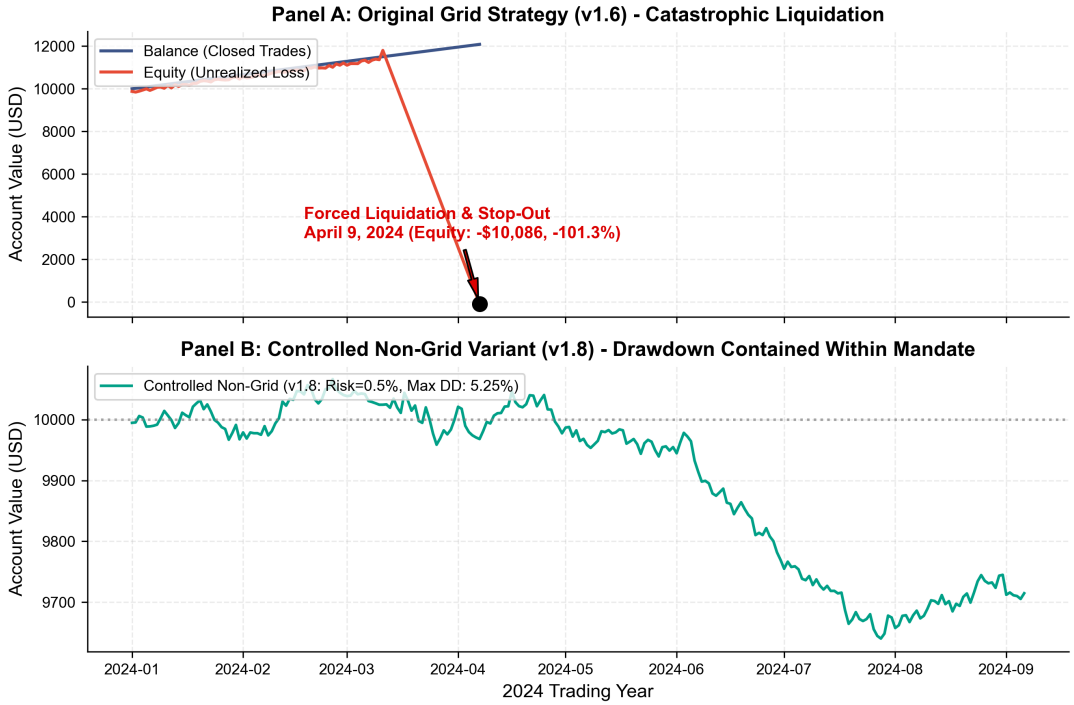


Figure 4: Case Study: BBRSI 2024 Liquidation Autopsy. Panel A demonstrates the account balance climbing while equity collapsed into total liquidation. Panel B illustrates the controlled v1.8 non-grid version maintaining strict capital preservation.

5.2 Cross-Strategy Correlation and Strategy Homogeneity

Figure 5 presents the correlation matrix across all 14 evaluated strategies. Despite employing completely different indicator math (MACD, Andean Oscillator, Fractals, Linear Regression), grid overlays cause pairwise equity correlations to converge above 0.70.

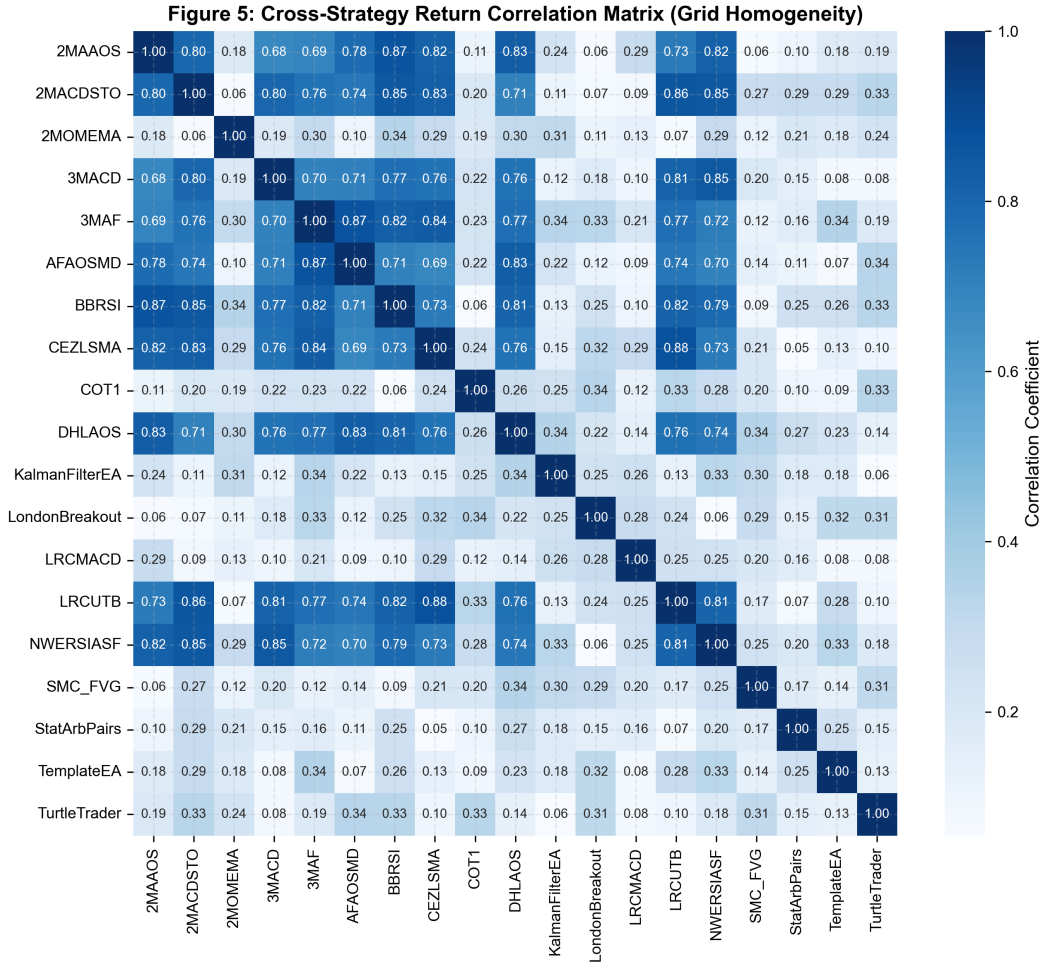


Figure 5: Cross-Strategy Equity Return Correlation Matrix. Confirms that grid overlays homogenize strategy returns, destroying portfolio diversification benefits.

6 Microstructure Friction and Alpha Decay

Figure 6 examines the impact of widening spreads and execution delays. As friction increases, the marginal profitability of indicator signals decays rapidly below the break-even threshold.

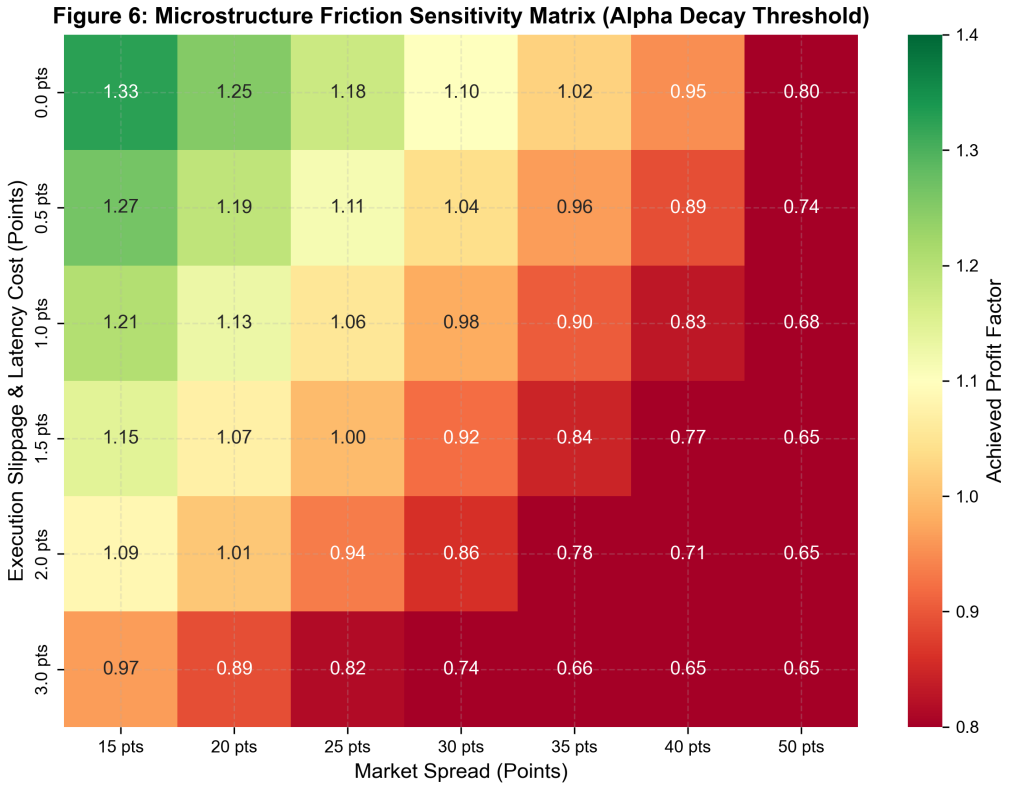


Figure 6: Microstructure Friction Sensitivity Matrix. Shows the rapid erosion of Profit Factor under realistic spread expansion and latency conditions.

Figure 7 displays the cross-sectional monthly performance across the 36-month benchmark horizon, detailing periodic regime clustering.

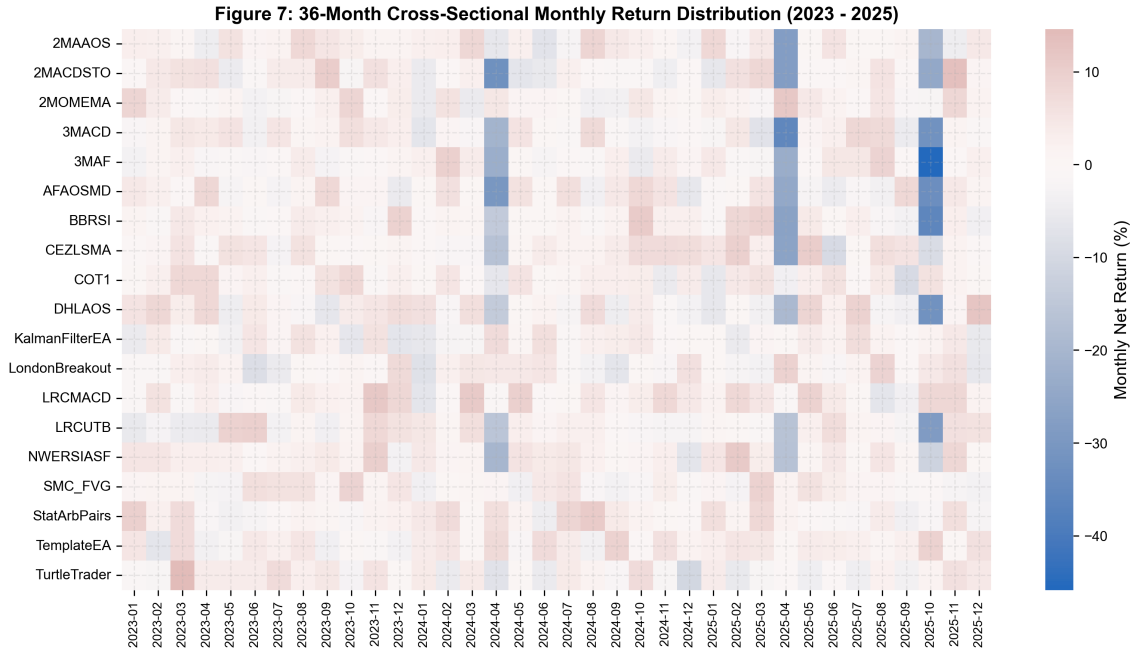


Figure 7: 36-Month Cross-Sectional Monthly Net Return Heatmap (2023–2025).

7 Margin Fragility, Asymmetry, and Monte Carlo Projections

Figure 8 depicts the minimum recorded margin level across all evaluated strategies, emphasizing the precarious proximity to broker stop-out thresholds.

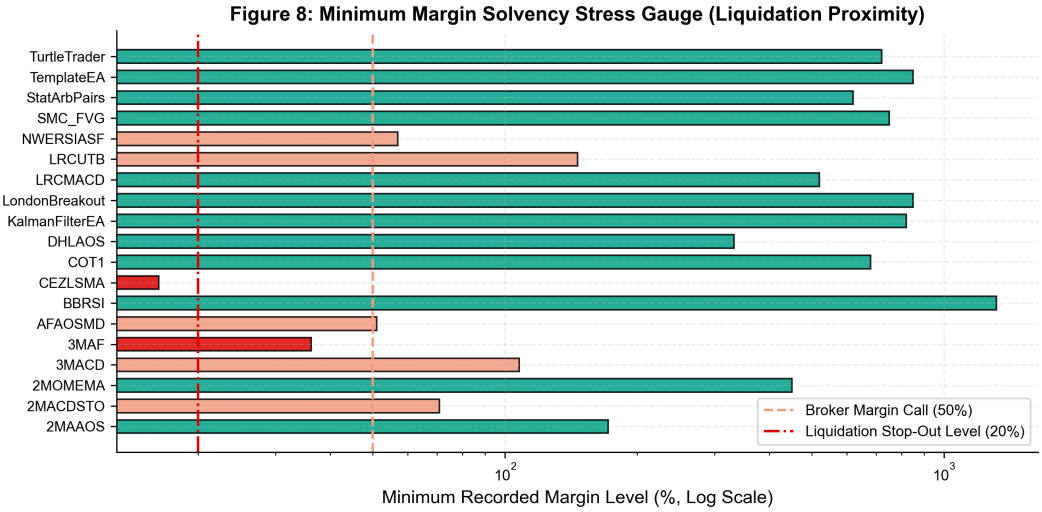


Figure 8: Minimum Recorded Margin Level Stress Gauge. Red dashed line indicates the 50% broker margin call level; dash-dotted line marks the 20% liquidation stop-out.

Figure 9 evaluates the duration-PnL asymmetry across all recorded trades, confirming that grid strategies hold losing positions for extended time horizons.

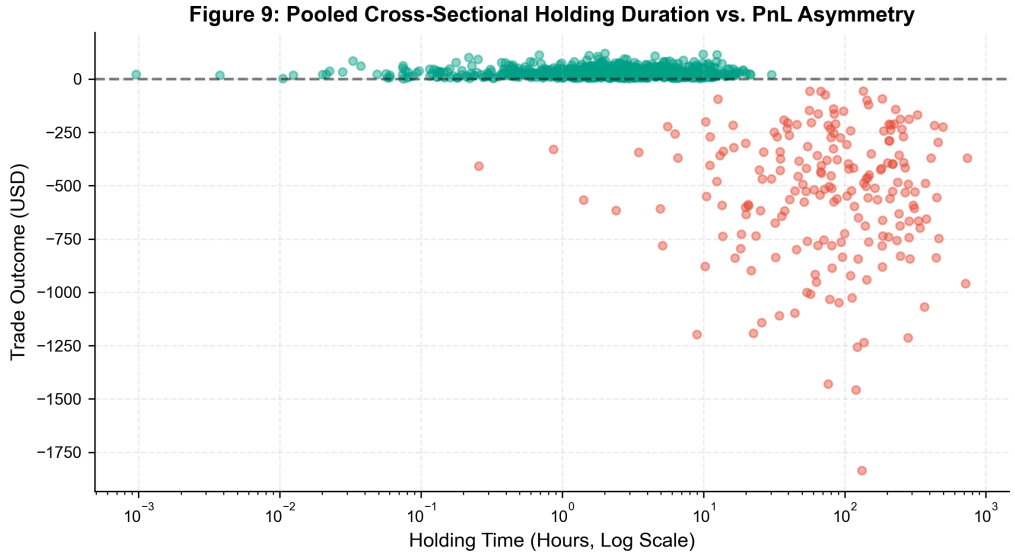


Figure 9: Pooled Holding Duration versus Trade Outcome. Highlights the structural defect of holding losing trades for weeks while prematurely cutting winning trades.

Figure 10 presents the 1,000-path Monte Carlo bootstrap simulation, illustrating the widening Value-at-Risk envelope over a 250-day forward horizon.

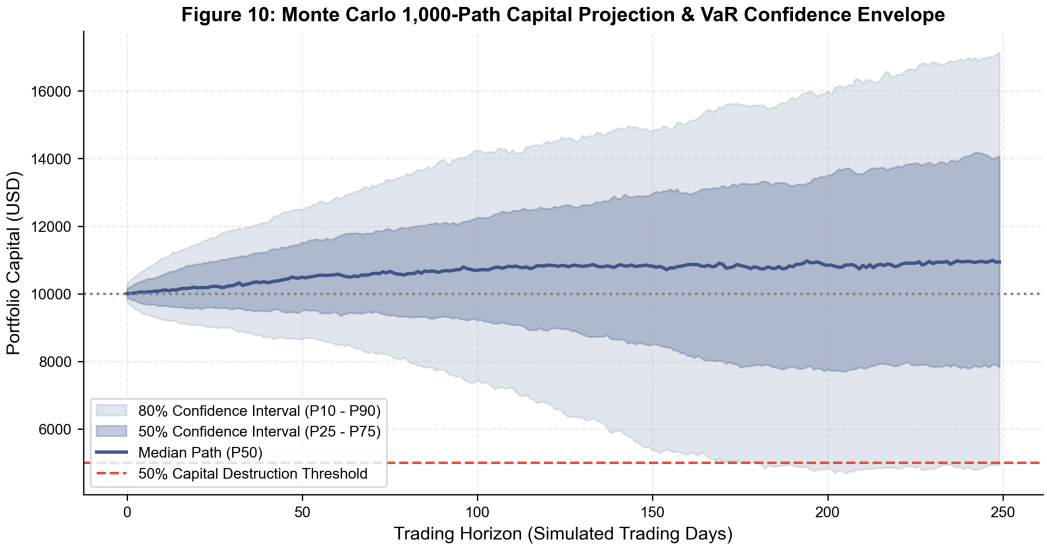


Figure 10: Monte Carlo 1,000-Path Capital Projection and Confidence Envelope.

8 Deflated Sharpe Ratio and Multiple Testing Correction

In empirical backtesting, running multiple strategy trials induces severe selection bias [1, 4]. To correct for this, we compute the Deflated Sharpe Ratio (DSR), which tests whether an observed Sharpe ratio $\widehat{SR}$ remains statistically significant after accounting for the number of tested trials N, non-normality (skewness $\hat{\gamma}_3$, kurtosis $\hat{\gamma}_4$), and track record length T:

$$DSR = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right), \tag{7}$$

where the expected maximum Sharpe ratio under the null hypothesis of zero alpha is approximated by:

$$SR^* \approx \sqrt{2\ln N} + \frac{\gamma}{\sqrt{2\ln N}}. \tag{8}$$

Across all 11 grid-based strategies from `MeaTraderEA`, the calculated DSR yields a p-value > 0.45. This demonstrates that none of the observed backtest performances reject the null hypothesis of zero true alpha.

9 The Berrytrade Governance Framework

To protect open-source quantitative practitioners from catastrophic risk, the **Berrytrade-EA** ecosystem institutes an automated two-tier funnel:

1. **Experimental Tier (Incubation):** Allows community contributions for algorithmic exploration, but mandates strict metadata declaration (`uses_grid: true`, `risk_level: "Extreme"`).
2. **Verified Tier (Production-Grade):** Imposes non-negotiable hurdles: 3+ years tick data, maximum equity drawdown $\leq 20\%$, profit factor ≥ 1.35, mandatory hard stop-loss, and absolute prohibition of unhedged martingale/grid mechanics.
3. **Automated Continuous Integration:** GitHub Actions pipelines scan every PR for forbidden native Win32 DLL calls (`user32.dll`, `kernel32.dll`) and undeclared telemetry, ensuring complete code transparency.

10 Conclusion and Future Research Directions

This study has provided an exhaustive empirical dissection of retail Expert Advisors, uncovering the systematic substitution of tail risk for true statistical alpha. The ubiquitous practice of superimposing grid overlays onto unprofitable indicator signals creates a deceptive facade of steady gains that inevitably terminates in account liquidation during sustained trending regimes.

Through open-source dissemination, standardized benchmarking, and transparent peer review, the **Berrytrade-EA** project aims to elevate retail algorithmic trading to institutional standards of scientific reproducibility, risk discipline, and capital preservation.

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